MLF Week-1 Solve With Us

September 27, 2025

$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$

$y = \beta_0 + \beta_1 x$

→ machine learning

In machine learning, Computers learn patterns from data and improve their performance on a task without being explicitly programmed with rules. Basically, "telling the computer what to do step by step," we give it data and let it discover the rules by itself.

Intuition: Traditional Programming: Input + Rules −− > $\boxed{\text{Output}}$
Machine Learning: $\boxed{\text{Input}}$ + Output (examples) −− > Algorithm finds the Rules

Example : Swiggy Delivery Time Prediction : Swiggy uses real-time traffic data, live weather updates, and historical delivery records to predict how long your order will take. By combining statistical models like $\boxed{\text{multiple regression}}$ with machine learning, they account for distance, road congestion, and even restaurant preparation speed - giving customers an accurate delivery estimate every time.

Another Examples:

- Banking: Detecting fraud in real-time.
- Social media: Recommending content.
- Healthcare: Predicting disease outbreaks from hospital records.

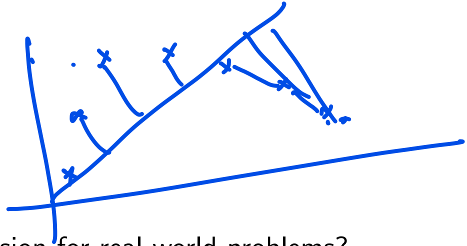
Q1

we can't make rule u wow".

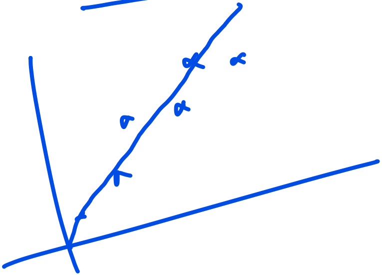
1. Suppose you are building a spam email filter. You have two options:
   - Write 200 rules manually to detect spam (e.g., if email contains "win money", or "lottery" etc.).
   - ✓ Feed 10,000 labeled emails to an algorithm to learn the rules.

   Which approach is Machine Learning and why?

Q2



Should we always use Linear Regression for real-world problems?



Q3

Arun goes for a walk on a daily basis and records the number of steps he has covered on each day using pedometer. The following table shows the recorded data for a week. He wants to know how many steps he can cover on the next day. Which ML model is more suitable here?

Reg / classification.

Day	Steps
05.09.2021	5800
06.09.2021	5945
07.09.2021	4880
08.09.2021	6120
09.09.2021	6430
10.09.2021	4640
11.09.2021	5980
12.09.2021	?

Regression

why

Predicting the continuous numerical value.

Q4

A behaviour analyst decided to study the emotional state of his spouse at the end of each day. He decided to observe and record various events (Features) that happen to her on a daily basis. The table below shows the data collected over a week. Which ML model would you suggest him?

	Day	shopping (x_1)	Housemaid Present (x_2)	walking (x_2)	Emotion (y)
1	05.09.2021	Yes 1	Yes 1	Yes 1	Happy → labeled
2	06.09.2021	Yes 1	No 0	Yes 1	Neutral → labeled
3	07.09.2021	Yes 1	No 0	No 0	Anger
4	08.09.2021	No	Yes	Yes	Happy
5	09.09.2021	No	Yes	No	Neutral
6	10.09.2021	Yes	No	No	Happy
7	11.09.2021	No	No	No	Anger
	12.09.2021	Yes	Yes	No	?

ordered rank | Anger < Neutral < Happy
1 2 3

Q5

Good negative

Which of the following are examples of unsupervised learning problems?

- (A) Grouping tweets based on topic similarity ✓
- (B) Grouping songs into playlists based on similarity. ✓
- (C) Checking whether an email is spam or not. → classification ✓
- (D) Identify the gender of online customers based on buying behaviour.

classification

input to output

A + 'A1'

B + 'B2'

F, M

Q6

Which of the following can form a good encoder decoder pair for d dimensional data ?

(A) $f : \mathbb{R}^d \to \mathbb{R}^{d'}$ where $d' < d$
$g : \mathbb{R}^{d'} \to \mathbb{R}^d$ where $d' > d$

(B) $f : \mathbb{R}^d \to \mathbb{R}^{d'}$ where $d' > d$
$g : \mathbb{R}^{d'} \to \mathbb{R}^d$ where $d' < d$

(C) $f : \mathbb{R}^d \to \mathbb{R}^{d'}$ where $d' < d$ ✓
$g : \mathbb{R}^{d'} \to \mathbb{R}^d$ where $d' < d$

(D) $f : \mathbb{R}^d \to \mathbb{R}^{d'}$ where $d' > d$
$g : \mathbb{R}^{d'} \to \mathbb{R}^d$ where $d' > d$

dimensionality reduction.

$d = 50$ features

without loosing too much information

$d' = 3$ features

$d = 50 \to d' = 60 \quad f(x) = \quad ; g(u).$

Q7

Consider the following data set where each data point consists of three features x_1, x_2 and x_3:

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Goal: $g(f(x^i)) \approx x^i$

$\|x\|^2 = x_1^2 + x_2^2 + x_3^2$

$\text{Loss} = \frac{1}{n} \sum_{i=1}^{n} \|g(f(x^i)) - x^i\|^2$

x_1	x_2	x_3
7	5	6
12	9	11
3	2	4
10	8	9

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

[redacted]

encoder decoder

76.5 Pair 1: $f(x_1, x_2, x_3) = 2x_1 - x_2 - x_3, \qquad g(u) = [u, u, u]$

2.93 Pair 2: $f(x_1, x_2, x_3) = \dfrac{x_1 + 2x_2 + x_3}{4}, \qquad g(u) = [u, u, u]$

Compute the reconstruction loss for each pair and identify which pair performs better.

Q7 Solution

$$\|g(f(x')) - x'\|^2$$

x_1	x_2	x_3	f_1	g_1	f_2	g_2
7	5	6	3	[3,3,3]	5.75	[5.75, 5.75, 5.75]
12	9	11	4	[4,4,4]	10.25	[10.25, 10.25, 10.25]
3	2	4	0	[0,0,0]	2.75	[2.75, 2.75, 2.75]
10	8	9	3	[3,3,3]	8.75	[8.75, 8.75, 8.75]

$$\text{loss(pair1)} = \frac{1}{4}\Big[\|[-4,-2,-3]\|^2 + \|[-8,-5,-7]\|^2 + \|[-3,-2,-4]\|^2 + \|[-7,-5,-6]\|^2\Big]$$

$$= \frac{1}{4}\Big[\big((-4)^2+(2)^2+(-3)^2\big) + \big((-8)^2+(5)^2+(7)^2\big) + \big((-3)^2+(2)^2+4^2\big) + \big(7^2+5^2+6^2\big)\Big]$$

$$= 76.5$$

Q7 Solution

$$\text{loss(pair2)} = \frac{1}{4}\sum_{i=1}^{4} \| g(f(x^i)) - x^i \|^2$$

$$= \frac{1}{4}\Big[\|[-1.25,\ 0.75,\ -0.25]\|^2 + \|[-1.75, 1.25, -0.75]\|^2 + \|[-0.25, 0.75, -1.25]\|^2 + \|[-1.25, 0.75, 0.25]\|^2 \Big]$$

$$= \frac{1}{4}\Big[(-1.25)^2 + 0.75^2 + (0.25)^2 + (-1.75)^2 + (1.25)^2 + (-0.75)^2 + (-0.25)^2 + (0.75)^2 + (-1.25)^2 + (-1.25)^2 + (0.75)^2 + (0.25)^2\Big]$$

$$= 2.93$$

Q7 Solution

$$\|x\|^2 = x^T x = [x_1 \;\; x_2 \;\; x_3]\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= x_1^2 + x_2^2 + x_3^2$$

Inner product $\langle x, x \rangle = \underline{\underline{x^T x}}$

Q8

The ML team in a movie production company wanted to predict whether a movie will earn 100 cr or not by using some classification models.
Therefore, the team collected presence/absence of various factors $x = [x_1, x_2, x_3]$ from the movies in the past. The data and the corresponding labels are shown in the Table below.

$(x^1, y^1), (x^2, y^2) \ldots (x^n, y^n)$

$\text{loss} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(f(x^i) \neq y^i)$

x	y
[0,1,1]	0
[1,0,1]	1
[1,0,0]	1
[1,1,1]	1
[1,1,0]	1

Compute the loss if the model predicts using $z = 0.5x_1 - 0.8x_2 + 0.4x_3$ and

$$u(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Q8 Solution

x	y	$z = 0.5x_1 - 0.8x_2 + 0.4x_3$	$u = \text{sign}(z)$
[0, 1, 1]	0	−0.4	0
[1, 0, 1]	1	0.9	1
[1, 0, 0]	1	0.5	1
[1, 1, 1]	1	0.1	1
[1, 1, 0]	1	−0.3	0

$$loss = \frac{1}{n} \sum_{i=1}^{n} 1\left(f(x^i) \neq y^i\right) = \frac{1}{5} = 0.2$$

Q8 Solution

Q9

$2.8 + 2$

Consider the dataset:

$$(x_i, y_i) = [(1,1),(2,2),(3,4),(4,5),(5,5)], \quad i = 1,2,3,4,5$$

and the regression model:

$$f(x) = x + 2$$

Compute the **mean squared loss** of $f(x)$ on this dataset.

$$\text{loss} = \frac{1}{n}\sum_{i=1}^{n}\left(f(x^i) - y^i\right)^2$$

Q9 Solution

(x_i, y_i): $(1,1), (2,2), (3,4), (4,5), (5,5)$

$f(x) = x + 2$

x	y	$f(x)$	$(f(x_i) - y_i)^2$
1	1	3	$(3-1)^2 = 4$
2	2	4	$(4-2)^2 = 4$
3	4	5	$(5-4)^2 = 1$
4	5	6	$(6-5)^2 = 1$
5	5	7	$(7-5)^2 = 4$

$$\text{Loss} = \frac{1}{5}(4+4+1+1+4) = \frac{14}{5} = 2.8$$

Ans

Thank You & Keep leaning!